		$T = 230\text{ N (2sf)}$
6	**D**	Since there is no net force on X and Y, $\Delta p_X = -\Delta p_Y \quad \cdots\cdots(1)$ $\dfrac{\Delta p_X}{\Delta p_Y} = \dfrac{m_X(-v_X - u_X)}{m_Y(v_Y - (-u_Y))} \quad \cdots\cdots(2)$ $-1 = \dfrac{-2.0}{3.0}\dfrac{(v_X + u_X)}{(v_Y + u_Y)}$ $= -\left(\dfrac{2}{3}\right)\dfrac{(v_X + u_X)}{(v_Y + u_Y)}$ $\dfrac{(v_X + u_X)}{(v_Y + u_Y)} = \dfrac{3}{2}$
